CROSS / PROBLEM 01

Problem 1: Higher-throughput encrypted MNIST on four TPU chips

CADathlon · `pset` · no AI assistance · ESSENTIAL problem · 10 points · ~1.5-hour topic within the proposed 8-hour event

The default short competition schedule is **student**: C3/C4, 100 images on one/four TPU chips, with a 300-second complete evaluation budget. Begin with `baseline_student.py`, then run `make -C src suite-evaluate-baseline`, `make -C src suite-evaluate`, and `make -C src suite-score` from this problem's directory after organizer asset preparation. The last command only scores existing protected reports; it does not run inference. Section 2.3.1 specifies this schedule. C5 and C6 (1000 images on one and four chips) are **hidden cases** of the same schedule: the organizer runs them after submission with the same ring, model and seed, and `make -C src suite-overall` folds them into the final 10-point score together with C3/C4. The original six-case workload remains an explicit `legacy` option and uses a separate calibration.

Like Problem 3, this problem is solved without generative AI or coding agents. The supplied source, tutorials, conventional editors, profilers, debuggers, documentation and deterministic autocomplete are permitted. P1 and P3 share one leaderboard.

1. Introduction

The FHE benchmarking suite of HomomorphicEncryption.org publishes a standardized ML-inference workload and a public results table. CROSS already runs that workload under CKKS on Google TPU v6e, correctly and at 128-bit security. It is not the fastest entry on that table.

Your task is to raise **throughput of the encrypted MNIST classifier across four TPU v6e chips** while preserving its outputs, weights and cryptographic security. Unlike Problem 3, which optimizes two CNNs each on one chip, this track is about **how many images per second a fixed serving stack delivers**, and the headroom is mostly in how images are placed into ciphertexts.

The legacy baseline leaves the ring nine-tenths empty. The local development command `make baseline-plan` reports that legacy plan:

```
{"depth": 3, "degree": 16384, "num_slots": 8192,
 "slots_used_by_input": 784, "slot_occupancy": 0.0957, ...}
```

One image occupies 784 of 8192 slots and the baseline puts one image in each ciphertext, so roughly 90% of every homomorphic operation is spent on zeros. The student baseline uses 4096 slots instead, with 784/4096 = 19.14% input occupancy; it still places one image per ciphertext. Published submissions to the same benchmark reach far lower per-image cost by placing the **batch** on the ring axis instead of the feature axis. Closing that gap is the core of this problem; it is a packing and scheduling problem, not a model problem.

2. Problem statement

2.1 The fixed network

Item	Value
Architecture	`Linear(784, 50)` → x^2 → `Linear(50, 10)`
Emitted depth	3
Data and client preprocessing	MNIST, 1×28×28; convert uint8 to [0,1], flatten to 784, then `(x-0.1307)/0.3081`
Client post-processing	argmax over the ten decrypted logits

The square is the activation: CKKS evaluates polynomials, and CROSS's activation registry accepts only a bare `x**(2**k)` squaring chain. The topology, widths, activation and preprocessing are fixed. Model weights are plaintext server constants; the privacy target is the client's image.

Organizers supply frozen trained weights and a hash manifest. The public fixtures in `test/` use seeded synthetic weights and are labelled so; they verify arithmetic, not classifier accuracy. Load **every** parameter from the official archive before a ranked run.

2.2 What to implement

Implement `build_service(model, input_shape, *, batch, devices)` in `src/bench_opt.py`. Return an object providing:

```
service.encrypt(images)   # (batch, 784) cleartext -> opaque ciphertext bundle
service.execute(bundle)   # ciphertext bundle -> ciphertext bundle
service.decrypt(bundle)   # ciphertext bundle -> (batch, 10) logits
service.release()         # free device memory
```

The bundle is deliberately opaque: it may be one ciphertext, a list, or any nesting of `Polynomial` values plus integer bookkeeping. **This is what lets you change the packing.** For student evaluation, preserve the `baseline_student.py` packing policy and supplied `keys` handling as your starting point. Copying the legacy default policy produces keys incompatible with the student suite. The baseline in `src/baseline.py` puts one image per ciphertext and one ciphertext per chip; you are not required to keep that.

`execute` runs on encrypted data only. It may not decrypt, call the cleartext model, inspect labels, cache by image identity or return precomputed answers. The local runner is a trusted test harness whose `Mapping` owns a context that also holds the secret key; evaluator code must not use that access. Include a source-level boundary audit in your report.

You may alter packing layout, plaintext-constant representation, BSGS planning, rotation reuse, kernel fusion, scheduling, device placement, memory reuse, batching across ciphertexts and exact modular kernels. The same optimization must work for arbitrary weights and inputs of the prescribed shapes. Edits to CROSS belong in `cross.patch`; edits under this problem's `src/` belong in the submission. You may add helpers. Do not change the benchmark, scoring, fixtures or checks.

You may not change weights, activation, input resolution, numerical preprocessing, topology, test data, ranked batch sizes or assigned devices. No pruning, retraining, approximate model replacement or plaintext fallback.

Cryptographic invariants. Keep a validated 128-bit `RingConfig`. The authored plan is depth 3, degree 16384, 8192 slots, 4 logical Q moduli, `dnum` 2, `log2(QP)` 365. A different secure ring is permitted **only** if `he_params.qp_is_secure` accepts it at 128-bit classical for the ring you actually run, and you print and report the derived parameters. Small insecure test rings receive no performance credit. Encryption noise must remain freshly sampled. Report `log2(Q)`, `log2(QP)`, the ceiling at your degree and the resulting margin.

Complete these tasks:

1. **P3.1 — establish the baseline:** verify provenance, run the cleartext and HE paths, and profile the baseline at each ranked batch on 1 and 4 chips.
2. **P3.2 — raise per-chip throughput:** attack slot occupancy or per-operation cost, and explain which operations or memory transfers it reduces.
3. **P3.3 — scale across chips:** show what your change does to 4-chip throughput and to scaling efficiency relative to one chip.
4. **P3.4 — compare and explain:** run C3/C4 for student, or all six ranked cases for explicit legacy evaluation; preserve numerical and label accuracy, and report throughput with evidence.

Suggested starting points: place several images in the slot axis of one ciphertext; batch along the ring dimension so the summation runs across ciphertexts and needs no rotations; reuse rotation keys; eliminate repacks; fuse; shorten buffer lifetimes; reduce the client codec cost, which dominates small batches. `Mapping.estimate_live_memory()` is a logical estimate; use a profiler for actual peak HBM and label the two separately.

2.3 Inputs, outputs and the legacy ranked workloads

The public `test1.in`–`test6.in` are JSON descriptions with batch size, device count, synthetic data seed and weight seed. `testN.out` holds real cleartext logits for every sample. The runner builds a deterministic weight set and checks shape, finite values, logit error and argmax agreement after HE. These fixtures use batches 1–8 on 1–4 devices to keep local development cheap; they are not the ranked suite.

All six ranked cases use TPU v6e chips from one host:

Case	Global batch	Chips	Mirrors
C1	1	1	single-image latency, one chip
C2	1	4	single-image latency, full host

Case	Global batch	Chips	Mirrors
C3	100	1	benchmark "small" instance, one chip
C4	100	4	benchmark "small" instance, full host
C5	1000	1	benchmark "medium" instance, one chip
C6	1000	4	benchmark "medium" instance, full host

C1 and C2 measure latency, where padding a batch of one across four chips cannot help. C3–C6 measure throughput, where packing can. A single change that helps only one regime is a partial result; say so in the report.

2.3.1 Short student schedule (default competition path)

For an approximately 1.5-hour topic slot, the `student` evaluation profile selects **C3 and C4** as its public cases: 100 images on one chip and on four chips. C5 and C6 (1000 images) are its organizer-run hidden cases, described below. C1 and C2 remain available for extended regression and latency experiments; the six local synthetic fixtures remain unchanged. This is a published workload change, not a faster result on the six-case legacy workload.

The student reference is `baseline_student.py`, with the same model and a 128-bit-secure N=8192 ring (42-bit composite scale, four HYBRID digits, log2(QP)=212.456 with ceiling 218). The suite supplies keys for that ring when `P1_BASELINE_PROFILE=student`. Candidate implementations must use compatible keys. Legacy baseline timings must not be used for this profile.

After staging the upstream suite, data and its cleartext checkpoint, measure the **complete** selected schedule with:

```
make -C src suite-evaluate-baseline
make -C src suite-evaluate
make -C src suite-score
```

The evaluation includes setup, three harness repetitions, per-case archiving, correctness checks and daemon cleanup. It fails if the entire schedule exceeds the budget. Baseline calibration and candidate results use distinct profile names. Inspect the generated report and logs; only actual TPU measurements can establish that a particular host meets five minutes. On the four-chip TPU v6e host (`v6e-4`, 2026-11-07) the complete baseline run of this two-case schedule finished in **276.63 seconds**, with three repetitions per case and 100% agreement with the ranked model in cleartext: C3 served 100 images in 1.8528 s on one chip (53.97 images/s) and C4 in 0.7680 s on four chips (130.21 images/s). See `cadathlon/validation/p1_student_suite.json` for measurements and limits; the earlier eight-chip run (257.47 seconds, C4 on eight chips) is kept in `cadathlon/`

`validation/p1_student_v6e8/report.json` as historical evidence. Recalibrate on your contest host. Dataset download and training the upstream harness's separate cleartext model are organizer asset preparation and must be completed before timing student evaluations.

The `student-v2` scorer requires a successful complete report with matching evaluation ID, C3/C4 archive hashes, frozen model/input identities, actual TPU hardware, software versions, unchanged sources, and completed cleanup within 300 seconds. A failed new attempt invalidates its previous latest-run report; reusing individual old successful cases cannot satisfy the complete-run gate. `suite-score` is offline arithmetic and returns nonzero for missing or stale evidence. CPU fixture checks and `suite-gates` do not satisfy these conditions.

Baseline publication stores only aggregate reports and measurement JSON under `src/suite/calibration/student-baseline/<evaluation_id>/`, bound by `baseline_manifest_student.json`. This proof ships with the release and survives `make clean`; it contains no per-image labels, predictions, pixels or keys. Failed recalibration preserves the prior published calibration. Candidate reports remain in `suite/run/` and must be regenerated after cleaning. The recorded 276.63-second four-chip run passed this complete `student-v2` contract and its baseline self-score (3.0 of 10 points: full correctness, no improvement points, not eligible for the ranking bonus). The earlier eight-chip 257.47-second `student-v2` run and the 262.14-second pre-v2 report are retained unchanged under `cadathlon/validation/p1_student_v6e8/report.json` and `cadathlon/validation/p1_student_pre_v2/historical_run.json`; the former was measured with C4 on eight chips and the latter predates device recording, so neither can replace a calibration of the four-chip schedule. The shipped proof also predates the hidden cases below: it is bound to the earlier `suite_config.json`, so `suite-score` rejects it until `make -C src suite-evaluate-baseline` has been run again on the contest host.

The 10-point event policy of section 4 is defined over the six ranked cases. If the short schedule is the one announced for an event, the same arithmetic applies over its cases. Scored alone, the two public cases carry 1.5 correctness points each, the improvement ladder is decided by the smaller of the two ratios, and the ranking bonus by their geometric mean (`score.py --evaluation-profile student`, or the Make target above). With the hidden cases below, the same arithmetic runs over four cases. Select one schedule, calibrate it on the contest hardware, and announce it before the event; do not compare scores or baselines across schedules/ring profiles.

Hidden cases C5 and C6 (organizer-run)

The student schedule has two **hidden cases**: C5 and C6, the benchmark's "medium" instance of **1000 images on one and on four chips**. They use the same `baseline_student.py` ring profile, the same frozen weights and the same seed 3. That seed selects 1000 MNIST test images for the medium instance, and the 100 images of C3/C4 are the first 100 of them. A correct `build_service` needs no change to run them; what they test is whether your packing and scheduling still hold when the batch is ten times larger and whether the four-chip scaling you report survives it. You do not run them before judging and their baseline throughput is not published; design for a general batch size and state in `REPORT.md` how your implementation scales from 100 to 1000 images.

The organizer uses one judge checkout on the contest host. It calibrates both schedules there once, then evaluates the submissions one at a time on both schedules in that same checkout: the scorer finds calibration proofs only under the checkout's own `src/suite/calibration/`, and the hidden calibration is git-ignored, so a fresh clone does not have it. To install a submission, copy in its `bench_opt.py` and any helper modules it adds under `src/`, and apply its `cross.patch`; the organizer's `src/suite/` files stay as they are. The hidden evaluation follows the same complete-evaluation rules as the public one (setup, three repetitions, archiving, correctness

checks and cleanup, bound to one evaluation ID, the frozen model, actual TPU hardware, unchanged sources and the software versions of the calibration) within its own budget, `hidden_budget_s` = 1800 seconds in `src/suite/suite_config.json`:

```
# once, in the judge checkout on the contest host
make -C src suite-evaluate-baseline          # public C3/C4 calibration
make -C src suite-evaluate-baseline-hidden   # hidden C5/C6 calibration
# per submission, after installing it in the same checkout
make -C src suite-evaluate                   # public C3/C4, rerun by the judge
make -C src suite-evaluate-hidden            # hidden C5/C6
make -C src suite-overall > /abs/teams/<team>/overall.json
```

`suite-overall` (`src/suite/overall.py`) scores both schedules itself (`make -C src suite-score-hidden` shows the hidden verdict alone) and re-applies the policy of section 4 over all four cases: the 3 correctness points are shared equally (0.75 per passing case), the improvement tier is decided by the **smallest** ratio over the four cases, and the ranking metric is the geometric mean of the four ratios. It refuses to combine a public and a hidden evaluation that were not produced by the same submission sources, frozen model and software versions. A missing, interrupted or failed hidden run fails C5 and C6 but does not void C3 and C4; because improvement points need every ranked case to pass, such a submission keeps only the 1.5 public correctness points and is not ranked. The next submission's evaluations replace these reports, so save each overall report before installing the next submission. `suite-rank` ranks the saved overall reports across teams unchanged.

2.4 Authored baseline observations

These were measured on the four-chip TPU v6e host (`v6e-4`, 2026-11-07) with the local runner (`runner.py --baseline`, two warmups, five measured requests) and the unmodified `baseline.py`. **They are indicative, not contest baselines**: the judge measures `T_base` on the official hardware with the frozen weights and images before the event, and those numbers govern scoring.

Measurement	Value
derived plan	depth 3, degree 16384, 8192 slots, 4 logical Q moduli, dnum 2
slot occupancy	784 / 8192 = **9.57%**
`log2(Q)` / `log2(QP)`	241.0 / 365.0, ceiling 438 at this degree, margin 73 bits
setup (build + keygen + compile), 1 chip	~85 s, paid by the first request; the constructor itself takes 0.5 s
end-to-end, batch 1, 1 chip	~142 ms, max abs logit error 2.4e-13
server execute only, batch 1, 1 chip	~29 ms
server execute only, batch 8, 4 chips	~63 ms for two lane groups of four, i.e. ~7.9 ms per image

Two things to read off this table before optimizing. First, at batch 1 the end-to-end time is roughly five times the server execute time, so the client codec, not the accelerator, dominates the latency cases. Second, the server time per lane group barely moves between one chip (about 29 ms for one image) and four chips (about 32 ms for four images), because every chip does the same mostly-empty work either way; the per-image gain comes only from the lane count.

2.5 The ranked evaluation runs inside the benchmark suite

The six ranked cases are not scored by the local runner in `src/`. They are scored by the **FHE benchmarking suite's own harness**, at the commit pinned in `src/suite/suite_config.json`. That is what makes the result deterministic and comparable with the public results table: correctness is decided against the suite's frozen ground-truth labels and performance comes from its `Timing` block. Neither is self-reported by the submission.

`src/suite/install_suite.py` writes eight small shims into the suite's `submissions/mnist/` directory. Each one loads your `bench_opt.py` through the same `build_service` entrypoint the local runner uses, so there is one implementation and two ways to exercise it. The installer **refuses to run if `harness/` differs from the pinned commit** — the suite's rules forbid changing the harness, and this enforces that mechanically rather than by trust.

It does change two files outside your submission directory, and you should know which. `harness/utils.py::build_submission` unconditionally runs `scripts/get_openfhe.sh` and `scripts/build_task.sh`, which for a pure-Python submission would download LibTorch, build OpenFHE, and then fail on a CMake project that no longer exists. The installer replaces both with no-ops and keeps the originals as `.orig`. That is outside `harness/`, so it is within the suite's rules, and `make -C src suite-check` prints exactly what it replaced.

The harness runs each stage as a separate process, which has three consequences you must design for:

1. **Client and server share one key pair.** `build_service` receives it as `keys`; pass it to `Mapping(..., keys=keys)`. A service that generates its own keys will encrypt under one key and decrypt under another.
2. **Your ciphertext bundle crosses a file boundary.** `src/suite/bundle_io.py` serializes any nesting of `Polynomial`, `int`, `float`, `None`, `list`, `tuple` and `dict`, preserving each ciphertext's tracked CKKS scale. You do not implement this, but a bundle containing anything else is rejected.
3. **Build lazily.** Only the evaluation process needs the compiled `Mapping`; the client processes need the codec. `baseline.py` builds the context on first encrypt and the `Mapping` on first execute for exactly this reason. The resident server forces both during stage 3, with a warm-up pass, so the measured stage never pays for compilation.

Determinism: the suite selects images from `--seed 3`, the organizer publishes the seed, and each case runs `--num_runs 3` with the median as the statistic. Read that last part precisely. `run_submission.py` re-creates its RNG from the base seed inside the run loop, so all three runs derive the same `generate_input` seed (`3` → `1742692731`) and evaluate **the same images**, despite a comment in the harness that says otherwise. The three runs are therefore repeat timings on identical input: they measure machine noise, not input variation. Do not read a tight spread across them as evidence that your implementation generalizes.

One more consequence of the harness's structure: stages 1, 2.2 and 3 run **once**, outside the run loop. Every `results-N.json` repeats the same `Key Generation` and `Encrypted model preprocessing` values, and every `Total` carries that one-time setup. Only stages 4–10 — including `Encrypted computation`, the stage your score is built on — are measured per run. Setup is gated separately, at 1800 s, and is not charged per image.

```
make -C src suite-check                          # verify the pinned checkout
make -C src suite-install                        # write the stage shims
make -C src suite-baseline SIZE=1 P3_DEVICES=4   # unmodified baseline, C4
make -C src suite-test     SIZE=1 P3_DEVICES=4   # your bench_opt.py, C4
make -C src suite-gates                          # gate self-test, no accelerator
make -C src suite-score                          # deterministic rubric arithmetic
make -C src suite-stop                           # release the accelerator
```

Each run is archived before the next one overwrites it. The suite keys its output by instance alone — `measurements/<instance>/` and `io/<instance>/` — so C1 and C2, which share the single instance and differ only in chip count, would otherwise overwrite each other. `suite/archive.py` copies each case's three `results-N.json` files under `src/suite/results/<case>_<label>/` and freezes the harness's correctness evidence alongside them. `suite-baseline` and `suite-test` invoke it for you.

One thing the suite **cannot** give you is the cleartext reference for the accuracy gate. `harness/cleartext_impl.py` runs `harness/mnist/mnist_ffnn_model.pth`, a 784-128-64-10 network, while this problem's fixed model is the 784-50-10 network with the `x**2` activation. The harness's `Harness model quality` is therefore a different network's accuracy, and the agreement between it and your encrypted output measures how differently the two were trained, not whether your HE pipeline is faithful. Gating on it would charge you for the harness's training run. `archive.py` instead runs **this problem's model in cleartext** on the harness's own pixel file and compares that, in organizer-owned code your `bench_opt.py` never touches. The measured baseline makes the distinction concrete: on the 100 small-instance images it scores 96/100 in cleartext and 96/100 encrypted — identical — while the harness's larger network scores 97/100. The gap to the harness model is a model difference; the gap to your own cleartext run, which is what section 4 gates, is zero.

Freezing that evidence is necessary because two things are absent from `results-N.json`. The single instance has **no `Quality` block**: the harness checks it with `verify_result` under `check=False`, so a FAIL neither stops the run nor is recorded anywhere in the measurement file. And `calculate_quality` divides the number of correct predictions by the number of predictions the submission wrote, so emitting fewer predictions than there are images yields a healthy-looking accuracy over the images that were skipped. `archive.py` recomputes both from the harness's own label and prediction files, and `score.py` refuses to award a case any points without that verdict. A case that writes an incomplete prediction set scores zero regardless of its speed.

`suite-score` then applies the policy of section 4 against the selected profile's calibration (`src/suite/baseline_manifest.json` with `P1_BASELINE_PROFILE=legacy`, `baseline_manifest_student.json` by default) and prints one JSON verdict. It is explicitly **not** an official score: the judge reruns on exclusive hardware with its own baseline manifest.

The unmodified `baseline.py` measured through this path, median of three runs, on the four-chip TPU v6e host (`v6e-4`, 2026-11-07; the complete six-case evaluation took 1685.9 s). `src/suite/baseline_manifest.json` holds these numbers; they are what `suite-score` divides by, and they are indicative until the judge measures its own. The earlier eight-chip legacy figures remain in this file's git history; `cadathlon/validation/p1_legacy_suite.json` records the four-chip run above.

case	images	chips	stage 7	img/s	ms/image	server execute	ms/image
C1	1	1	0.1193 s	8.38	119.3	0.0284 s	28.4

case	images	chips	stage 7	img/s	ms/image	server execute	ms/image
C2	1	4	0.1221 s	8.19	122.1	0.0306 s	30.6
C3	100	1	3.0700 s	32.57	30.7	2.6976 s	27.0
C4	100	4	1.1557 s	86.53	11.6	0.7533 s	7.5
C5	1000	1	30.0190 s	33.31	30.0	26.9654 s	27.0
C6	1000	4	10.7333 s	93.17	10.7	7.5094 s	7.5

Three things in that table are the assignment. On one chip the server cost is flat at **27.0 ms per image** whether the batch is 100 or 1000, because the baseline walks the batch one lane at a time — it is sequential by construction. Four chips buy **3.59×** at batch 1000 (26.97 s → 7.51 s), not 4×. And at batch 1 four chips are slower than one (30.6 ms against 28.4 ms), because a single image cannot fill four lanes and the cross-device work is pure overhead. Combine that with the 9.57% slot occupancy from section 2.4: the baseline leaves most of the ring and most of the machine idle.

Correctness is not the thing to optimize here — it is already exact. Across C3–C6 the encrypted output agreed with the same model in cleartext on **every one of the 1100 images**, and `own_cleartext_agreement` is 1.0 in all six cases. Your job is to keep that while making the table's right-hand columns smaller.

For orientation only, the suite ships the reference implementation's own measurements at the pinned commit, generated in February 2026 and copied to `src/suite/results/_upstream_reference/`. Median `Encrypted computation`:

instance	images	compute (s)	ms/image	accuracy
single	1	7.6145	7614.50	not reported for this instance
small	100	647.0703	6470.70	0.9600
medium	1000	6505.6093	6505.61	0.9740

Those runs were made on the reference implementation's machine, which the repository does not describe, so the gap to the CROSS baseline mixes hardware with implementation and is **not** a measured speedup. Your score is computed against the CROSS baseline on contest hardware, never against this table.

Organizers supply 64 public calibration images and select 256 disjoint hidden test images, with 25 or 26 per class. Identical images and weights are used for the baseline and each candidate. Evaluate every image in a ranked batch without padding or dropping samples; where the batch exceeds 256 images, the organizer publishes the repeat schedule. Public boundary cases additionally include all zeros, uniform values, high-contrast pixels and distinct images in every lane. The harness may change seeds and weights for structural tests to catch specialization; ranked accuracy always uses the frozen trained weights.

For **each** case all logits must satisfy `abs(HE−clear) <= 0.02 + 0.01*abs(clear)`, at least **99%** of image argmaxes must match the cleartext reference, and label accuracy may fall by at most **1 percentage point** relative to that same model's cleartext predictions on the same images. "That same model" is literal and matters: it is this

problem's 784-50-10 network in cleartext, not the harness's 784-128-64-10 reference, and not the harness's `Harness model quality` figure. On the suite path `archive.py` computes it; the logit gate applies on the local runner, which is the only path that sees cleartext logits rather than labels. Report absolute accuracy too. Require all three gates. NaN, Inf, a shape mismatch or a dropped image fails the case. Organizers validate that the unmodified baseline passes before release. Tolerances are contest choices, not guarantees from CKKS.

A note on precision if you change the ring: reducing the modulus is a legitimate optimization, but the logit error grows quickly as the chain shortens. The authored plan decrypts within about 2e-12 of cleartext; a 100-bit chain at degree 4096 measured about 2e-03 on the same network family. The gates above are absolute, so verify them at your chosen parameters rather than assuming the baseline's margin carries over.

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3. How to compile and run

The organizer image contains Python, NumPy, PyTorch, JAX, the TPU runtime and the pinned CROSS checkout. `CROSS_ROOT` selects another checkout if needed. The six public reference tests run without training or downloading.

```
cd cadathlon/pset/p1_benchmark/src
make all
make reference-test      # numerical fixtures only, no encryption
make baseline-plan       # actual CROSS plan and slot occupancy, no keygen
make baseline-test       # unmodified encrypted CROSS, test1; minutes/GB
make test                # all six cases using your bench_opt.py
python runner.py --case ../test/test6.in --baseline
make submit              # writes ../submission.tar.gz
```

The local commands above are for development. The ranked cases run through the benchmark suite (section 2.5):

```
cd cadathlon/pset/p1_benchmark
make -C src suite-check                        # pinned checkout, harness intact
make -C src suite-install
make -C src suite-baseline SIZE=2 P3_DEVICES=4 P1_BASELINE_PROFILE=legacy
make -C src suite-test     SIZE=2 P3_DEVICES=4 P1_BASELINE_PROFILE=legacy
make -C src suite-gates
make -C src suite-score P1_BASELINE_PROFILE=legacy
make -C src suite-stop
```

`SIZE` selects the suite instance (0 single, 1 small, 2 medium) and `P3_DEVICES` the chip count; together they pick the ranked case. `MLINF_ROOT` points at the suite checkout if it is not `~/ml-inference`.

The editable TODO deliberately raises `NotImplementedError` until completed. `baseline.py` provides the legacy reference; `baseline_student.py` is the default competition starting point. `make all` checks Python syntax and runs the pure arithmetic scorer self-tests; it performs no TPU evaluation. Logs go to stderr and one machine-readable JSON result goes to stdout per case. Do not byte-compare noisy CKKS outputs to `.out`; use the stated bounds.

Legacy per-case runtime limits: up to four TPU v6e chips on the fixed host image, 512 GiB maximum host RSS, and no device OOM. Prepare, key generation and compilation may take at most **30 minutes per configuration**. Each online batch may take at most **300 seconds**. A failed or timed-out case scores zero. CPU runs are useful for correctness but do not count toward the TPU score.

The student schedule additionally requires **the complete C3/C4 evaluation within 300 seconds**, including setup, repetitions, validation and cleanup. The longer legacy per-case limits do not override that whole-run budget.

Prepare once per configuration and release it before building the next. The local synthetic runner defaults to two warmups and five measured requests on the same fixture, with fresh encryption every time. Its server-only and client-to-client times are development diagnostics, not the ranked metric.

The ranked suite performs stage-3 warmup once, then **three** harness runs on the same seed-derived images, as explained in section 2.5. Ranked throughput is `X = B / median(Timing["Encrypted computation"])` : the harness's own stage 7, including its subprocess/RPC and ciphertext loading/storing work. It does not include client encryption/decryption. Synchronize encrypted outputs before stopping the server timer. Also report server-only time and full evaluation wall time separately. Baseline and candidate use identical timing boundaries, images and assigned device counts.

Compilation, key generation and constant preparation are reported separately and excluded from warm `T`. They still obey the setup limit. Reuse only public weights, constants, keys and executables, never image-specific ciphertext intermediates. The judge runs baseline and candidate in separate clean processes, interleaves their reservations and reruns ties within 2% once. Record physical peak device memory and process RSS, plus logical estimates if used. Throughput obtained by changing the ranked batch or the assigned chip count does not count as an optimization.

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4. How to score

Problem 1 is worth **10 points** in the event, scored by policy `p1-10-point-v1` in three tiers. Let `G_i` equal 1 only if ranked case i passes all numerical, accuracy, privacy, parameter, memory and runtime gates, otherwise 0, and let `R_i = X_i / X_base_i` be its throughput ratio against the organizer's calibrated baseline on the same host and schedule, using organizer-rerun medians. The policy is defined over the six ranked cases C1–C6; section 2.3.1 says how it maps onto the short schedule, alone (C3/C4) or with its hidden cases (C3/C4 + C5/C6, combined by `suite-overall`).

Tier 1 — correctness, 3 points. The 3 points are shared equally by the ranked cases: `3 * sum_i G_i / 6`, so each passing case earns 0.5 points. A faithful, unoptimized baseline scores exactly 3.

Tier 2 — improvement, up to 4 points. Only a submission that passes **all** ranked cases earns improvement points, and its smallest ratio decides the tier, `R_min = min_i R_i` :

`R_min`	meaning	points
≥ 1.05	at least 5% more images per second on every case	1
≥ 1.10	at least 10% on every case	2
≥ 1.20	at least 20% on every case	3
≥ 1.40	at least 40% on every case	4

A ratio exactly on a threshold qualifies. One case below 1.05 caps the tier at 0 however large the other speedups, and one failed case removes the tier entirely; the latency cases C1/C2 therefore count as much as the throughput cases.

Tier 3 — ranking bonus, up to 3 points. Submissions that earned at least one improvement point are ranked by the geometric mean of their six ratios, `(prod_i R_i)^(1/6)`, rounded to four decimals. The best receives 3 bonus points, the second 2 and the third 1. A k-way tie on the rounded metric shares the higher bonus and skips the following k−1 ranks; bonuses are not redistributed when fewer than three teams are eligible. Submissions without an improvement point are not ranked: this eligibility gate is an organizer decision, so that a regression on any case cannot be outweighed by speedups elsewhere.

A team's total is `correctness + improvement + bonus`, at most 10. `make -C src suite-score` prints the first two tiers and the ranking inputs for one submission; `make -C src suite-rank RANK_REPORTS="team=report.json ..."` (`src/suite/rank.py`) assigns the bonus across the saved reports of all teams. Both are deterministic arithmetic over the suite's own measurement files, and neither is an official score without the judge's rerun. The judge's recipe is:

```
# per team, every team on the same schedule and the same calibration manifest
make -C src suite-score P1_BASELINE_PROFILE=legacy > /abs/teams/<team>/score.json
# once, across teams; absolute paths, make's echo lines in the saved files are tolerated
make -C src suite-rank RANK_REPORTS="<team>=/abs/teams/<team>/score.json ..."

# student schedule with hidden cases: score public and hidden runs together, then rank
make -C src suite-overall > /abs/teams/<team>/overall.json
make -C src suite-rank RANK_REPORTS="<team>=/abs/teams/<team>/overall.json ..."
```

Each report records the SHA-256 of the calibration manifest it was scored against; `suite-rank` refuses to mix reports from different schedules, policies or manifests, lists tied teams alphabetically and never redistributes an unclaimed bonus.

`X_i` comes from the suite's own `results-{1,2,3}.json`, archived per case under `src/suite/results/`, never from a contestant-printed number. The accuracy and argmax gates come from `archive.py`, which recomputes them from the harness's label and prediction files and from this problem's model run in cleartext — organizer-owned code, not the submission's. `src/suite/score.py` covers the numerical, accuracy, completeness, setup and runtime gates. The privacy, parameter and memory gates in `G_i` are assessed by the judge from `REPORT.md` and a source audit; a case that fails any of them scores zero regardless of what the scorer printed. The report earns no separate points under this policy, but a submission cannot pass that audit without it: state

checkpoint hashes and the environment, baseline profiles at 1 and 4 chips with slot occupancy before and after, the measured bottlenecks your change addresses, complete raw timing and actual-memory records, regression tests, and the privacy and parameter review with `log2(QP)`, the ceiling and the margin.

The judge runs baseline and candidate in separate clean processes, interleaves their reservations, and reruns submissions whose ranking metrics lie within 2% of each other once before assigning the bonus; metrics still equal after rounding share the bonus as described above. For the published leaderboard order only, remaining ties break by lower peak device memory, then lower total setup time, which the judge applies by hand to the alphabetical order printed by `suite-rank`.

Submit `src/`, `cross.patch`, `REPORT.md`, raw result JSON and environment and weight hashes. Include both successful and failed configurations. Four report pages are sufficient; logs and code are additional. Do not submit datasets, trained weights, secret keys, caches, generated reference outputs or unrelated files. `make submit` packages the `src/` sources, `cross.patch`, `REPORT.md` and every `results*.json` in this problem directory, so copy each raw result JSON here under that name first (reports under `src/suite/run/` and `src/suite/results/` are not packaged). It does not upload anything.

5. References

1. Tong et al., Leveraging ASIC AI Chips for Homomorphic Encryption, HPCA 2026; supplied `CROSS_paper.pdf` and paper. Use BAT/MAT and profiling ideas; paper numbers are not contest baselines.
2. FHE benchmarking suite, ML-inference workload: repository and published results. The results table shows what other submissions achieve on this workload and on what hardware; treat those numbers as context, not as contest baselines.
3. CROSS `jaxite_word/API_REFERENCE.md`, `nn.py`, `packing.py`, `mapping.py`, `bsgs.py`, `he_params.py`, `profiler.py`; `profile_analysis/analyze_trace_json.py`. `API_REFERENCE.md` documents the canonical rank-5 `Polynomial`, `global_batch`, device placement and the codec boundary.
4. CROSS tutorial: CROSS, TPU Novice to Master, NTT Algorithms and Open Challenge. Current canonical code takes precedence over older insecure-ring performance tables.